

CHAUTAUQUA COUNTY AP, NY, USA

WMO: 725235

Lat: 42.150N

Lon: 79.250W

Elev: 1723

StdP: 13.80

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04720

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	4.5	-7.7	3.9	2.0	-2.3	5.2	6.9	25.5	25.3	23.4	24.7	7.7	240	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	82.3	69.7	81.1	68.6	78.9	67.0	72.4	79.7	70.8	77.6	69.2	75.7	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	118.5	76.9	68.3	111.0	74.8	66.3	103.4	73.1	37.2	79.4	35.7	77.5	34.3	76.3	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	18.9	17.2	DB	-5.1	86.7	7.2	3.4	-10.3	89.1	-14.5	91.2	-18.5	93.1	-23.7	95.6	
			WB	-5.7	75.5	7.1	1.7	-10.8	76.7	-14.9	77.7	-18.9	78.7	-24.0	79.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	22.8	24.8	32.6	44.8	55.9	63.9	67.7	66.2	60.2	49.6	38.0	29.1
	DBStd	18.16	11.37	11.20	11.33	9.77	8.50	6.49	5.07	5.01	6.91	8.43	9.35	9.17
	HDD50	3502	845	706	553	214	38	1	0	5	114	375	651	
	HDD65	7100	1308	1125	1004	608	301	96	31	47	175	480	811	1114
	CDD50	2197	1	1	14	58	222	418	549	503	312	102	15	2
	CDD65	322	0	0	0	3	20	63	115	85	32	4	0	0
	CDH74	1797	0	0	1	27	138	390	645	406	178	12	0	0
	CDH80	256	0	0	0	1	12	57	108	54	24	0	0	0
	WSAvg	8.2	9.9	9.2	9.1	9.1	7.8	7.1	6.5	6.1	6.6	8.0	9.2	9.6
	PrecAvg	43.3	3.1	2.4	2.7	3.5	3.6	3.7	4.4	3.9	4.3	4.1	3.7	3.7
Precipitation	PrecMax	54.2	5.7	5.2	4.6	6.1	8.2	7.8	8.7	6.9	7.8	6.9	5.4	6.5
	PrecMin	34.4	1.6	1.0	1.5	1.9	1.3	0.6	1.6	2.2	1.7	2.0	1.6	2.3
	PrecStd	5.3	1.1	0.9	0.9	1.1	1.7	1.6	1.6	1.2	1.4	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	57.0	70.4	78.6	81.7	84.3	87.2	84.1	82.4	75.1	64.7	58.6
		MCWB	52.9	48.6	57.5	60.2	66.5	71.1	72.3	70.5	69.3	64.1	54.6	53.2
	2%	DB	50.6	51.6	62.6	71.9	78.7	81.6	82.3	81.4	79.2	71.2	61.0	52.0
		MCWB	48.3	46.3	51.1	56.9	64.2	69.0	70.6	69.3	67.9	61.1	52.8	48.7
	5%	DB	44.5	45.7	55.2	65.9	74.5	79.2	80.9	79.0	74.9	66.3	56.6	46.2
		MCWB	41.6	40.5	47.7	53.8	62.4	67.2	69.1	67.6	65.4	59.2	50.4	42.9
	10%	DB	38.2	40.5	49.8	61.1	70.4	75.2	78.4	75.3	72.2	63.1	53.6	42.0
		MCWB	35.2	35.4	43.1	50.3	59.4	64.7	67.5	66.1	64.0	56.6	48.5	38.3
	0.4%	WB	53.3	51.3	58.2	62.7	69.1	73.1	75.2	73.8	72.8	67.9	57.5	54.3
		MCDB	56.5	55.0	68.3	74.2	78.0	81.4	82.8	81.2	79.3	72.8	62.4	57.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.3	45.9	53.1	58.9	66.8	70.8	72.9	71.3	69.5	64.1	54.4	48.8
		MCDB	49.9	49.1	58.9	68.2	75.5	79.2	80.4	77.6	76.6	67.6	58.6	50.8
	5%	WB	41.8	40.8	48.4	55.5	64.1	68.9	70.9	69.8	67.4	60.4	51.2	43.7
		MCDB	44.1	45.5	55.1	64.1	71.5	76.6	77.6	75.6	73.0	65.2	55.6	46.3
	10%	WB	35.6	36.0	43.6	52.2	61.5	67.1	69.1	68.1	65.3	57.0	47.8	38.5
		MCDB	38.2	40.2	50.6	59.1	68.4	73.7	75.3	74.0	69.9	62.1	52.5	41.5
Mean Daily Temperature Range	5% DB	MDBR	12.4	14.2	15.0	18.0	17.6	16.6	16.3	15.9	16.4	14.8	12.3	10.4
		MCDBR	17.0	19.4	22.1	25.2	21.3	19.8	18.6	18.8	20.0	20.0	18.7	16.1
		MCWBR	15.5	15.8	14.7	15.6	11.8	10.6	9.6	10.0	10.7	12.0	13.8	13.9
	5% WB	MCDBR	16.8	19.0	20.2	22.2	19.1	17.2	16.4	15.8	17.1	17.3	16.5	15.8
		MCWBR	16.2	16.6	14.7	15.6	11.8	10.6	9.4	9.6	10.2	11.9	14.1	14.8
Clear-Sky Solar Irradiance	taub		0.292	0.322	0.323	0.352	0.383	0.414	0.415	0.406	0.357	0.343	0.302	0.289
	taud		2.401	2.299	2.406	2.386	2.325	2.262	2.277	2.291	2.410	2.449	2.534	2.470
	Ebn at Noon		271	278	293	291	283	273	271	270	278	268	268	263
	Edh at Noon		25	32	33	36	40	43	42	40	33	28	22	22
All-Sky Solar Radiation	RadAvg		421	680	1012	1394	1682	1831	1854	1652	1312	814	512	325
	RadStd		38	77	123	149	170	146	143	95	130	84	75	42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A

Nomenclature: See separate page